

Early Detection of ALL Disease Using YOLOv4 Algorithm on Peripheral Blood Smear Images

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Abstract—Peripheral smear is a method that allows the analysis of blood with the help of a microscope after staining the blood liquid on a glass slide. The morphological appearance and multiplicity of some blood elements are examined by this method. This examination is preferred for evaluating the response to treatment as well as providing information about the diagnosis and severity of diseases. Since manual examination of blood elements by operators contains many humanistic and environmental parameters, it is time-consuming and error-prone. Instead of this approach which does not offer a standard accuracy rate, it is planned to create a quick, cost-effective and fully automatic computer-aided system. In this study, the examination of ALL (Acute Lymphoblastic Leukaemia) malignancy which is one of the main types of leukaemia by peripheral smear method will be handled. ALL is formed by the overproduction of malignant and immature white blood cells called lymphoblasts. Since it spreads to the blood circulation and vital organs rapidly, the speed of the tests for early diagnosis is important. The peripheral blood smear test, which is a part of the diagnosis process of ALL malignancy, is a guide in terms of diagnosis and severity of the disease with the presence and multiplicity of lymphoblast cells. Therefore, a fully automatic system is designed to determine the presence and number of blasts on the given image. The model used to create this system is trained with the YOLOv4 algorithm. Training is provided via peripheral blood smear images the free and public in the ALL-IDB dataset (<https://homes.di.unimi.it/scotti/all/>) tagged by expert oncologists. As a result of the test process, the success rate, precision, recall, specificity and F score criteria are obtained as 98.87%, 99.08%, 99.81%, 88.88% and 99.44%, respectively. With this work, it is real-time given the total number of blast cells on the image, which is an important biomarker in determining the severity of ALL disease or evaluating the response to treatment. It is expected that this fully automatic system which produces output after 3-4 seconds for the given input will be a helpful idea for the operators.

Keywords—*ALL malignancy, peripheral blood smear, YOLOv4 algorithm*

I. INTRODUCTION

Acute lymphoblastic leukaemia is a hematic disorder known as cancer of the white blood cells. It is occurred by the excessive production of malignant and immature white blood cells known as lymphoblast or blast produced in the bone marrow. This malignancy, which generally affects young children and adults over 50, spreads rapidly to the bloodstream and vital organs. Therefore, early diagnosis is the basic principle for the sustainability of vitality[1].

In the first stage of ALL cancer, general symptoms such as weakness, weight loss, loss of appetite, nosebleeds, bone or joint pains and shortness of breath are observed. However, in the later stages, blast cells that are overproduced in the bone

marrow begin to be seen in the peripheral blood. Evaluation of peripheral blood film is part of the diagnostic process[1].

Examination of the blood film by means of a microscope is a method that provides qualitative and quantitative information. Operators provide an evaluation in two ways. The first analysis is based on cell morphology, which provides inferences related to tumour pathologies and differentiated structures. The second analysis is based on the number of white blood cells. But, the manual analysis and counting of cells as manual depend on the information and skill of the expert. This situation affects the final result. On the other hand, an expert can't analyze and count tens of thousands of cells on a blood slide without being distracted[2]. Therefore, manual analysis does not offer a standardized accuracy rate as it does not enable an automatic analysis. It is also error-prone and time-consuming. [2][3]. Cytometry technology is available in the market for these challenges. But, this technology used in counting cells is unsuccessful as it destroys blood samples during evaluation. Also, it is an expensive technology [2][4].

Automatic detection and counting of blood cells contain important information in the framework of diseases related to viability. Therefore, in the scope of the efficiency principle, it is aimed to realize a technology that enables automatic analysis of blood elements and produces fast and objective results[5]. For this purpose, different many computer-assisted studies have been carried out from the past to the present. [6] performed classification of 6 different white blood cells in peripheral blood smear and [7] distinguished eight different blood cells in peripheral blood. [8] and [9] classified WBCs into five different categories by field adaptation-transfer learning and capsule networks, respectively. [10] provided WBC, RBC, and platelet analysis on the peripheral blood smear which contains different important information about the person's health status. [11] performed automatic recognition and classification of leukocytes in peripheral blood to assist operators. [5] developed a fully automatic algorithm for the identification of WBCs. [12] proposed the WBC nucleus segmentation approach for the medical diagnosis of white blood cells which have an important role in the immune system. [13] presented a study to distinguish between red blood cells and white blood cells. [14] carried out a model for nucleus and cytoplasm segmentation of WBCs in the framework of various approaches. [15] used the fuzzy logic algorithm to detect WBC and RBC structures accurately. [16] proposed a semi-automatic study to provide white blood cell segmentation. [17] carried out a study to detect leukaemia disease from the number of blood cells. [18] proposed a segmentation approach based on the nuclei and cytoplasm elements of white blood cells. [19] presented a

structure using spectral and spatial features to separate lymphoblasts from lymphocytes.

This study is planned to perform the analysis of blood cells in a fast and cost-effective manner. And also, it is aimed to design a fully automatic system that assists the operators in obtaining the standard accuracy rate. The flow diagram of the current study focusing on the separation and counting of immature white blood cells in the blood from other elements is presented in Figure 1.

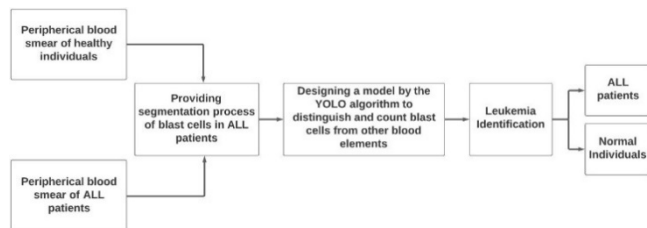


Figure 1: Flow chart of the presented study

Blast cells observed in the peripheral blood smear are identified by the segmentation process. A model is designed using the YOLO algorithm to distinguish the blast cells from other blood elements on the image. As a result of the training process, the classification performed on the image of the peripheral blood smear is provided fully automatically.

II. METHODOLOGY

A. Dataset

In the present study, it is planned to distinguish ALL patients from healthy individuals. For this purpose, analyzes are made on the microscopic images of peripheral blood samples obtained from the website <https://homes.di.unimi.it/scotti/all/>. All images in the public and free ALL-IDB dataset are captured at different magnifications and densities using the Canon PowerShot G5 camera. Assessments are provided using images from the ALL-DIB1+ALL-IDB2 datasets.

B. Segmentation Process

Peripheral smear used to accelerate the diagnosis process, provide early cancer diagnosis, and contribute positively to the medical world's treatment processes is a crucial haematology test for individuals with ALL diseases. But, since it contains many humanistic and environmental disadvantages, it does not offer a standard accuracy rate; it is slow and error-prone. Therefore, it is planned to make a fully automatic system with the multidisciplinary studies of the informatics and medical world. However, the segmentation process, which plays an essential role in recognising blast cells as a critical biomarker for ALL detection, is a necessary step for the correct diagnosis and treatment process. The trained model's reliability will come to the fore with the accurate segment of the blast cells. For this purpose, the segmentation process of the images is carried out using the <https://www.makesense.ai/> site. Then a .zip package

containing files in YOLO format is downloaded. Thus, the necessary segmenting step for learning is provided.

C. YOLO(You Only look once) Approach

YOLO is a deep learning network that detects and classifies blast cells in blood samples captured by the microscope. The inputs given to the algorithm are the data obtained with different magnification rates from areas with varied densities of blood elements. Version 4 of Yolo is preferred for accurate distinguishing and counting of blast cells segmented on these data[20].

The YOLOV4 algorithm developed by [21] is an enhanced version of YOLOV3. The backbone part is the CSPDarknet53 structure, in which CSPNet and Darknet53 architectures are used together. This structure which plays a role in feature extraction reduces computation and memory costs and improves the learning ability of CNN[22]. The neck part of the YOLOV4 algorithm consists of SPP and PAN approaches and the head part consists of the YOLOV3 algorithm[21][23].

SPP is a module that improves the receptive field and consists of Spatial Pyramid Matching (SPM). SPM (Spatial Pyramid Matching) divides the input image as SXS into a grid to obtain the spatial pyramid. It is then integrated into CNN by SPP. The PAN structure is used to collect parameters from different backbone levels[21][23].

YOLOV4 being a one-step method produces output by calculating the coordinates and category probability of the given input. The target detection process is presented in Figure 2[24].

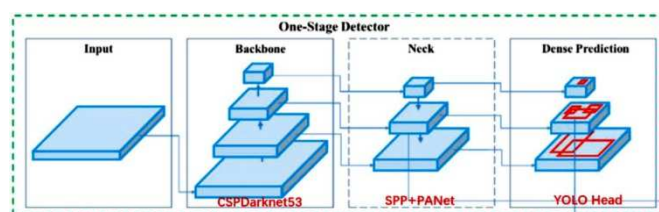


Figure 2. Target detection process for YOLOv4 algorithm [24]

The YOLO algorithm, in which the image is considered as a whole, separates the data into SXS coordinates. It estimates the bounding box coordinates for the relevant object in the image and evaluates the probability that the object in these boxes is a blast cell. The probability of whether a blast cell is present or not is available for all grids, and aspect ratios are estimated based on the entire image[25].

The rates of loss value obtained as a result of the training performed on the ALL-IDB1+ALL-IDB2 datasets with the YOLOV4 algorithm are presented in Figure3.

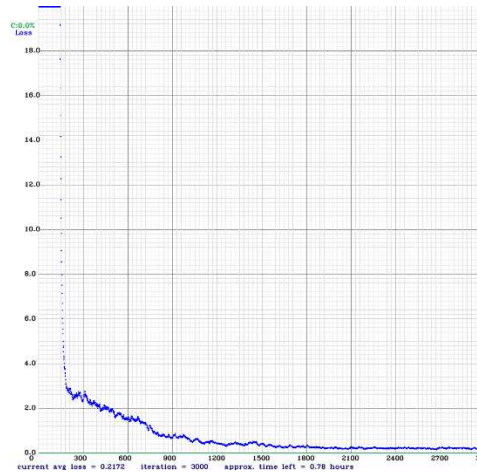


Figure 3. Detection and counting process on ALL-IDB1+ALL-IDB2 dataset

When the graph is examined, it is seen that the loss value is reached a stable state as a result of the learning process performed on ALL-IDB1 + ALL-IDB2 datasets. As a result of the test process, the success rate, precision, recall, specificity and F score criteria are obtained as 98.87%, 99.08%, 99.81%, 88.88% and 99.44%, respectively. This points out that the trained model will provide highly clear inferences on different images that it has never seen.

III. DISCUSSION AND RESULTS

Peripheral blood smear used in the discovery of ALL disease is the preferred method of analysis to examine blood elements. This analysis contains changes like colour, shape, size and amount of blood elements. This process is provided by manual examination by doctors. But, manual evaluations on the slide using a microscope contain many parameters that will adversely affect the final result. These parameters can be caused by environmental factors such as the contrast or brightness of the image, while they also can be caused by humanistic factors such as the operator's carelessness, intensity, fatigue or knowledge level. Therefore, in the scope of the efficiency principle, it is aimed to carry out a fully automatic study that is fast, successful, cost-effective and provides standard accuracy.

In this context, training is conducted with images in which ALL-IDB1 + ALL-IDB2 datasets are used together. It is observed that the obtained results produce competitive outputs with low loss value and high recognition rate in a short time without preprocessing step. At the same time, automatic specification of the total number of blast cells on the image which is an important biomarker in determining the severity of ALL disease or evaluating the response to treatment is carried out in real-time.

The results produced by the sample images given to the trained model are presented in Figure 4 and Figure 5.

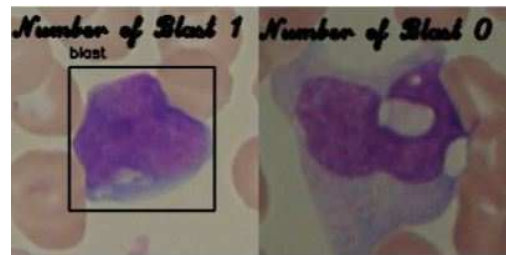


Figure 4. Detection and counting process on ALL-IDB1 dataset

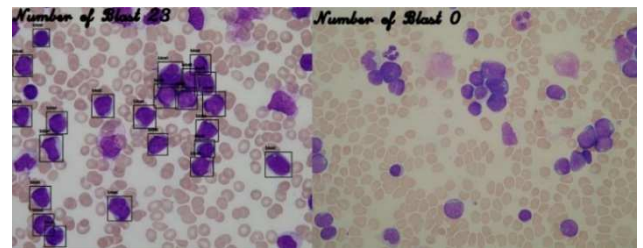


Figure 5. Detection and counting process on ALL-IDB2 dataset

In the presented study, blast cells in digitized peripheral blood smear images are detected in real-time. This process is carried out with the YOLOv4 approach using the ALL-IDB dataset. Some of the similar studies published in the last 3 years using the ALL-IDB dataset are given in Table 1.

Compared to the studies in the table, the highlight feature of this study is the real-time presentation of the number of blast cells, which is a biomarker in expressing the severity of ALL disease. This cell count output would eliminate the need for Flow Cytometry technology, which is an expensive technology and is not available in many places.

Thus, It is expected that the real-time automatic system created with the output produced after 3-4 seconds for the given input will be a useful tool for the operators.

TABLE1: A COMPARISON ON STUDIES USING THE ALL-IDB DATASET

Tablo1	A comparison on studies using the ALL-IDB dataset				
	Author	Aim of study	Method	Dataset	Results
	Jha, K. K. & Dutta, H. S. 2019[26].	Leukaemia detection modul from the blood smear images	MI+LDP+SCA based deep CN classifier	ALL-IDB2	Accuracy is %98,7.
	Al-jaboriy, S. S. Sjarif, N. N. A. Chuprat, S. & Abdualлах, W. M. 2019[27].	To develop a new and improved technique for leukocyte cell segmentation	GA+ANN	ALL-IDB1	Identified blast cell accuracy is %97
	Al-Tahhan, F. E. Fares, M. E. Sakr, A. A. & Aladle, D. A. 2020 [28].	To identify automatically the ALL subtypes(L1/L2/L3)	KNN/SVM/ANN	ALL-IDB2	Maximum accuracy is obtained as %100 by quadratic SVM
	Safuan, S. N. M. Tomari, M. R. M. Zakaria, W. N. W. Mohd, M. N. H. & Suriani, N. S. 2020[29].	To differentiate lymphoblast and non-lymphoblast cells and examine ALL disease	VGG/GoogleNet/Alexnet	ALL-IDB2(60 images)	Maximum accuracy is obtained as %99,13 by VGG
	Sahlol, A. T. Kollmannsberger, P. & Ewees, A. A. 2020[30].	White blood cell leukaemia image classification	SESSA+VGGNet	ALL-IDB2	Accuracy is %96,11.
	Anilkumar, K. K. Manoj, V. J. & Sagi, T. M. 2021[31].	Automated identification of leukaemia	Transfer Learning Architectures	ALL-IDB1 ALL-IDB2	Maximum accuracy is %100 both ALL-D-IDB1 and ALL-IDB2 dataset
	Khandekar, R. Shastry, P. Jaishankar, S. Faust, O. & Sampathila, N. 2021[32].	Automated blast cell detection for early diagnosis of leukaemia	Preprocessing + Augmentation + YOLOV4 algorithm with 416x416 size of YOLO image	ALL-IDB1	Mean average Precision is %95,57. Recall is %92. F1 Score is %92.
	Proposed Approach	Full-automated blast cell detection and counting for early diagnosis of leukaemia	YOLOV4 algorithm with 512x512 size of YOLO image	ALL-IDB1 + ALL-IDB2	Accuracy rate is %98,87. Precision is %99,08. Recall is %99,81. Specificity is %88,88. F score is %99,44.

IV. CONCLUSION AND FUTURE WORKS

This study aims to evaluate ALL disease using the peripheral blood smear technique. As a result of the study, the output will be produced in 3-4 seconds with real-time automatic diagnosis provided on digitized peripheral blood smear images. With this study on the identification and counting of blast cells from hundreds of cells, the severity of the disease will be evaluated, the right treatment protocol will be selected and the response to the treatment will be monitored. It is expected that this study will provide helpful ideas to physicians.

In the future, this approach can be integrated into an android-based system. It is expected that the real-time application created with the mobile application offering a simple interface can spread in the medical world in a short time.

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